

Siddhartha Gairola

PhD Student in Computer Vision & Machine Learning
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Google Scholar 
Saarbrücken, Germany 

Education

MPI for Informatics & Saarland University

Ph.D. Student, Computer Science

Saarbrücken, Germany

Oct 2022 – present

- Advisors: Prof. Bernt Schiele, Prof. Francesco Locatello

Institute of Science and Technology Austria (ISTA)

Visiting Researcher, Causal Learning and Artificial Intelligence Lab

Klosterneuburg, Austria

Aug 2024 – Feb 2025

International Institute of Information Technology, Hyderabad

MS by Research in CSE

Hyderabad, India

2018 – 2020

- Advisor: Prof. P. J. Narayanan
- Thesis: *Image Representations for Style Retrieval, Recognition and Background Replacement Tasks*
- CGPA: **9.75/10**

BTech with Honours in CSE

2014 – 2018

- CGPA: **8.91/10**

Work Experience

Microsoft Research, India

Research Fellow

Bengaluru, India

Aug 2020 – Aug 2022

- Advisors: Dr. Mohit Jain, Dr. Nipun Kwatra
- Developed a low-cost smartphone-based corneal topographer for keratoconus detection in collaboration with Sankara Eye Hospital. Published at ACM IMWUT/Ubicomp 2021.
- Worked on training DNN models for abnormality detection in limited data settings. Published at EMBC 2021.

Research Intern

Jan 2020 – Jul 2020

- Advisors: Dr. Mohit Jain, Dr. Nipun Kwatra
- Worked on building smartphone based diagnostic solutions for eye diseases using mobile computing, computer vision and machine learning.

Adobe Systems, India

Product Intern, Media and Data Science Research Lab

Noida, India

Jun 2019 – Jan 2020

- Advisors: Balaji K., Mayur Hemani
- Worked on Guided Few-shot Image Segmentation, improved on the prior state-of-the-art by 6%.
- Published at IJCAI 2020 and filed a patent.

IIIT Hyderabad

Research Assistant, Center for Visual Information Technology (CVIT)

Hyderabad, India

May 2016 – May 2019

- Advisor: Prof. P. J. Narayanan
- Worked on representation learning for image style search and retrieval. Published at WACV 2020.
- Also worked on the task of color-consistent background replacement. Published at MMM 2018.

Google Summer of Code (GSoC)

Student Developer

Scilab, remote

May 2018 – Aug 2018

- Accepted as a developer for the Google Summer of Code Program for the 2nd time with Scilab.
- Implemented a demo in C/C++ and Scilab as a working example for the MEX Library in Scilab.

Student Developer

May 2017 – Aug 2017

- Accepted as a developer for the Google Summer of Code Program with the organization Scilab.
- Implemented a C/C++ wrapper for Matlab MEX-API on current API Scilab.

Patents

- S. Gairola, M. Hemani, A. Chopra, J. Dahl, Balaji K. Improved Similarity Propagation for One-Shot and Few-Shot Image Segmentation (US 16/906,954), 2020.

Peer-Reviewed Publications

- 14 S. Gairola*, A. Pardy*, S. Rao, A. Wróbel, B. Zieliński, B. Schiele, D. Rymarczyk. **DisParQ: Self-Supervised Part Concepts for Interpretable Vision Foundation Models**. arXiv preprint arXiv:2610.09802, 2026.
- 13 S. Gairola, J. Xie, A. Kukleva, F. Locatello, B. Schiele. **Stranger Things: When Objects Appear Without Their Typical Neighbours**. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- 12 S. Gairola*, A. Wróbel*, J. Tabor, B. Schiele, B. Zieliński, D. Rymarczyk. **DAVE: Distribution-aware Attribution via ViT Gradient Decomposition**. *International Conference on Machine Learning (ICML)*, 2026. ★ Spotlight, top 2.2% of ~24K submissions.
- 11 S. Gairola*, R. Maser*, S. Rao, B. Schiele. **Align Once to Explain: Feature Alignment for Scalable B-cosification of Foundational Vision Transformers**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- 10 S. Gairola, M. Böhle, F. Locatello, B. Schiele. **How to Probe: Simple Yet Effective Techniques for Improving Post-hoc Explanations**. *International Conference on Learning Representations (ICLR)*, 2025.
- 9 V. Ganatra, S. Gairola, P. Joshi, A. Balasubramaniam, K. Murali, A. Varadharajan, B. Mallikarjuna, N. Kwatra, M. Jain. **SmartKC++: Improving Performance of Smartphone-Based Corneal Topographers**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025.
- 8 S. Gairola, P. Joshi, A. Balasubramaniam, K. Murali, N. Kwatra, M. Jain. **Keratoconus Classifier for Smartphone-based Corneal Topographer**. *IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2022.
- 7 A. Aggarwal, S. Gairola, U. Upadhyay, A. P. Vasishtha, D. Rao, A. Goyal, K. Murali, N. Kwatra, M. Jain. **Towards Automating Retinoscopy for Refractive Error Diagnosis**. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 6(3), 2022.
- 6 S. Gairola, M. Bohra, N. Shaheer, N. Jayaprakash, P. Joshi, A. Balasubramaniam, K. Murali, N. Kwatra, M. Jain. **SmartKC: Smartphone-based Corneal Topographer for Keratoconus Detection**. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 5(4), 2021.
- 5 S. Gairola, F. Tom, N. Kwatra, M. Jain. **RespireNet: A Deep Neural Network for Accurately Detecting Abnormal Lung Sounds in Limited Data Setting**. *IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2021.
- 4 S. Gairola, A. Chopra, M. Hemani, Balaji K. **SimPropNet: Improved Similarity Propagation for Few-Shot Image Segmentation**. *International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.
- 3 S. Gairola, R. Shah, P. J. Narayanan. **Unsupervised Image Style Embeddings for Retrieval and Recognition Tasks**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2020.
- 2 V. Kumar*, D. Khattar*, S. Gairola*, Y. K. Lal*, V. Varma. **Identifying Clickbait: A Multi-Strategy Approach Using Neural Networks**. *ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2018.
- 1 S. Rawat*, S. Gairola*, R. Shah, P. J. Narayanan. **Find Me a Sky: A Data-Driven Method for Color-Consistent Sky Search and Replacement**. *International Conference on Multimedia Modeling (MMM)*, 2018.

*equal contribution

Academic Service

Organizer Computer Vision for Developing Countries (CV4DC) Workshop at ACCV 2024, ICCV 2025

Reviewer ICML 2023 2024 2025 2026 ★ NeurIPS 2022 2023 2025 ICLR 2022 2023 2024

CVPR 2024 2025 2026 ECCV 2024 ICCV 2025 IHCI 2021

★ Gold reviewer

Volunteer ICVGIP 2018

Talks & Conference Presentations

- Intriguing Applications and Overlooked Pitfalls of XAI in Visual Models, at the GMUM Workshop, Jagiellonian University, October 2024.
- A DNN for Accurately Detecting Abnormal Lung Sounds in Limited Data Setting, EMBC 2021 (video).

- Improved Similarity Propagation for Few-shot Image Segmentation, *IJCAI 2020* ([video](#)).
- Unsupervised Image Style Embeddings for Retrieval and Recognition Tasks, *WACV 2020* ([video](#)).

Selected Research Projects

Smartphone-based Corneal Topographer

Nov 2020 – Oct 2021

Developed a low-cost smartphone-based corneal topographer. It provides a portable and scalable solution for the mass screening of keratoconus. Built the entire system: an AI powered smartphone app, the 3D printed attachment, and the processing software. [*Android, Python, PyTorch*; [project page](#)]

Abnormality Detection in Respiratory Signals

Aug 2020 – Nov 2020

Developed a 'ResNet34'-based model along with a suite of novel augmentation and fine-tuning techniques to effectively utilize small-sized datasets. The method improved upon the state-of-the-art for 4-class classification on the ICBHI dataset by 2.2%. [*Python, PyTorch*; [code](#)]

Similarity Propagation for Few-Shot Image Segmentation

Jun 2019 – Jan 2020

Developed a deep learning system that leverages background-foreground similarity to perform few-shot image segmentation. The method achieved state-of-the-art results on the PASCAL-5i, COCO-20i and FSS datasets. [*Python, PyTorch*]

Unsupervised Image Style Representation Learning

Jan 2019 – Jun 2019

Developed a deep learning model to learn robust style representations without supervision. Achieved state-of-the-art results across six datasets and it was published as a paper. [*Python, PyTorch*; [code](#)]

Sketching with Style

Aug 2018 – Dec 2018

Implemented the ICCV 2017 paper "Sketching with Style" ([link](#)). Reproduced the results provided in the paper and open-sourced the code. [*PyTorch, Python*; [code](#)]

Neural Clickbait Detection Engine

May 2017 – Aug 2017

Developed a system to robustly identify clickbait social media posts. The system comprises a siamese network that fuses visual and textual information to learn a neural embedding used to detect clickbait. [*Python, Keras*; [code](#)]

Color Consistent Background Replacement

Jan 2017 – Apr 2017

Developed a data-driven method for color-consistent sky search and replacement. A diverse set of skies consistent in color and illumination are retrieved from a curated dataset and used to generate composites, which are re-ranked based on realism and diversity. [*MATLAB*; [project page](#)]

Teaching Experience

 **Saarland University** · Elements of Data Science and Artificial Intelligence (Winter 2023, 2024, 2025, 2026)

 **IIIT Hyderabad** · Computer Graphics (Spring 2019), Digital Image Processing (Monsoon 2017, 2018), Computer Vision (Spring 2018), Artificial Intelligence (Spring 2017), Digital Logic and Processors (Monsoon 2016)

- Duties as a TA involved taking regular tutorials, setting up assignments and conducting evaluations.

Awards & Achievements

 **Dean's Research Award** · for a paper published at an international conference while an undergraduate 2018

 **Dean's List Award** · six consecutive semesters: Monsoon '15, '16, '17 and Spring '16, '17, '18 2015 – 2018

 **ACM ICPC Asia Onsite Regionals** · qualified twice 2015, 2016

 **JEE Advanced** · 98.5 percentile among 150,000 candidates 2014

 **JEE Main** · 99.7 percentile among 1.2 million candidates 2014

Technical Skills

Working Knowledge	Bash, C/C++, Python, MATLAB, Octave, OpenCV, Android-SDK
Software & Tools	GNU/Linux, HTML, CSS, JavaScript, LaTeX, Excel
Deep Learning Frameworks	PyTorch, TensorFlow, Keras, Caffe
Past Experience	Java, OpenGL, WebGL, Django, Web2py